

the gap this document closes

Writing Proofs, from Zero

A 22-cell notebook, explained — with the
method drawn from eight books on your shelf

Devlin · Diedrichs & Lovett · Stewart & Tall · Tao

Halmos · O’Leary · Radożycki · Thurston

2,164 pages, of which you read 134.

```
ns=3: True
ns=5: True
ns=8: True
```

↓
is not
↓

$$\forall N \in \mathbb{N}$$

6

proof techniques,
each fully worked

9

steps for *finding*
a proof (Tao)

22

notebook cells,
each explained

1

trap: a CAS
that says False

The one idea. Your Sage notebook prints True three times and it feels like proof. It is not. Devlin’s own book gives the killer example: **the Goldbach Conjecture has been verified for every even number up to 1.6 quintillion and remains unproved.**

Checking is a *search*. Proving is an *argument*. This document teaches the six arguments you need, then assembles them into one real theorem about the partition function you are already computing.

Contents

ten sections · six techniques · twenty-two cells · eight books

1	How to use these three things together	books → this PDF → notebook
2	What a proof is — three definitions, from three of your books	Devlin · D&L 2.1.15 · Thurston
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4	How to <i>find</i> a proof — Tao’s nine steps	and how §9’s theorem was actually found
5	The six techniques, cell by cell	CELL 7 CELL 9 CELL 11 CELL 13 CELL 15 CELL 17
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7	The payoff — the theorem, assembled	$\text{Tr}(T^N) = \lambda_+^N + \lambda_-^N$, for every N
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The three artefacts this document sits between

the eight books	Partition_Function_Study/Maths_Proof_Skill/ — 2,164 pp, 22 MB, MD5-verified
this document	Partition_Function_Study/ — the map between them
the notebook	Proof_Techniques_01_Sage_10.9_From_Pure_to_Applied.ipynb — 22 cells, Sage 10.9, 0 errors

1 How to use these three things together

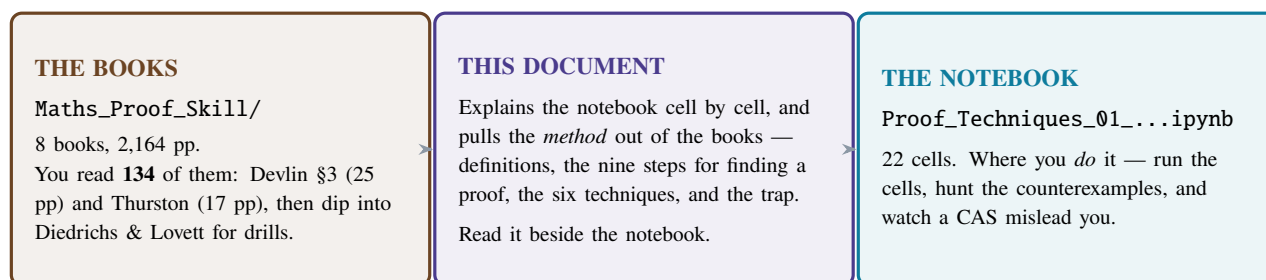


Figure 1. The books supply the method, this document supplies the map, the notebook is where you practise. None of the three works well alone.

2 What a proof is — three definitions, from three of your books

Devlin, *Introduction to Mathematical Thinking*, §3.1 (p. 117 book, 01_Start_Here)

*“Proofs are constructed for two main purposes: to establish truth and to communicate to others. . . a proof of a statement must explain **why** that statement is true.”*

Note the second purpose. A proof that convinces only you is half a proof.

Diedrichs & Lovett, *Transition to Advanced Mathematics*, Definition 2.1.15 (02_Main_Reference)

*“A proof of a mathematical statement is a **sound argument** that establishes the truth of the conclusion from the truth of the stated hypotheses.”*

Sharper and more formal. Notice what it does *not* say: nothing about examples, computation or evidence.

Thurston, *On Proof and Progress in Mathematics*, p. 2 (01_Start_Here)

*“How do mathematicians advance human **understanding** of mathematics? . . . what we are doing is finding ways for people to understand and think about mathematics.”*

Thurston reframes the whole activity. He then gives the four-colour theorem as his example: it was proved by massive computation, and the controversy

*“had little to do with doubt people had as to the veracity of the theorem or the correctness of the proof. Rather, it reflected a continuing desire for **human understanding** of a proof, in addition to knowledge that the theorem is true.”*

That is your situation exactly, every time Sage prints True.

Putting the three together

A proof is

1. a **sound argument** (Diedrichs & Lovett) — each step follows by a rule of logic from definitions, axioms, or theorems already proved;
2. that **explains why** (Devlin) — not merely that;
3. and **communicates** that explanation to a reader (Devlin, Thurston).

A computation can satisfy none of these and still be correct. That is the whole difficulty.

3 The evidence trap — and why your notebook opens with it

Cells 1–2 of the notebook are logic. But §0, the very first markdown cell, exists to break one habit before it forms.

The habit: mistaking verification for proof

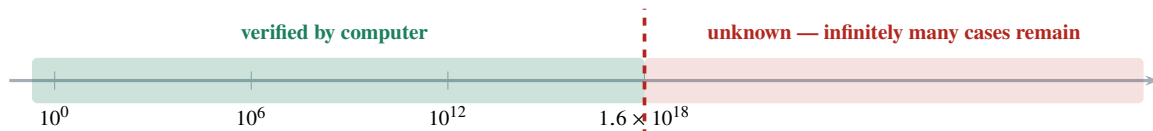
Your companion notebook prints

```
ns=3: Tr(T^ns) == lam_+^ns + lam_-^ns -> True
ns=5: ... True
ns=8: ... True
```

Three Trues feel conclusive. They are three facts about three numbers. The theorem claims something about **every** N , of which there are infinitely many.

Devlin's killer example, §3.1 — read this before you ever trust a pattern

*“Proving a mathematical statement is much more than gathering evidence in its favor. . . the Goldbach Conjecture. . . It is possible to run computer programs to check the statement for many specific even numbers, and to date (July 2012) it has been verified for all numbers up to and beyond 1.6×10^{18} (1.6 quintillion). Most mathematicians believe it to be true. **But it has not yet been proved.**”*



Goldbach's conjecture. No finite amount of green closes the red. The same picture applies to $ns=3, 5, 8$.

Diedrichs & Lovett name the error, §2.3.6

*“Try a Few Examples: Sometimes, running examples with the objects in question allows us to discern patterns. . . The patterns generally do not constitute a proof, so simply claiming ‘This pattern continues’ is a form of **faulty generalization**.”*

So examples are *encouraged* — as a way to find the proof. They are just never the proof.

The one sentence to carry

*Checking is a search. Proving is an argument.
A search over an infinite set never terminates.*

4 How to find a proof — Tao's nine steps

The techniques in §5 tell you what a finished proof looks like. They do not tell you how to get one. For that your shelf has Terence Tao's *Solving Mathematical Problems* (04_Problem_Strategy), whose Chapter 1 lays out a sequence you can actually follow when stuck.

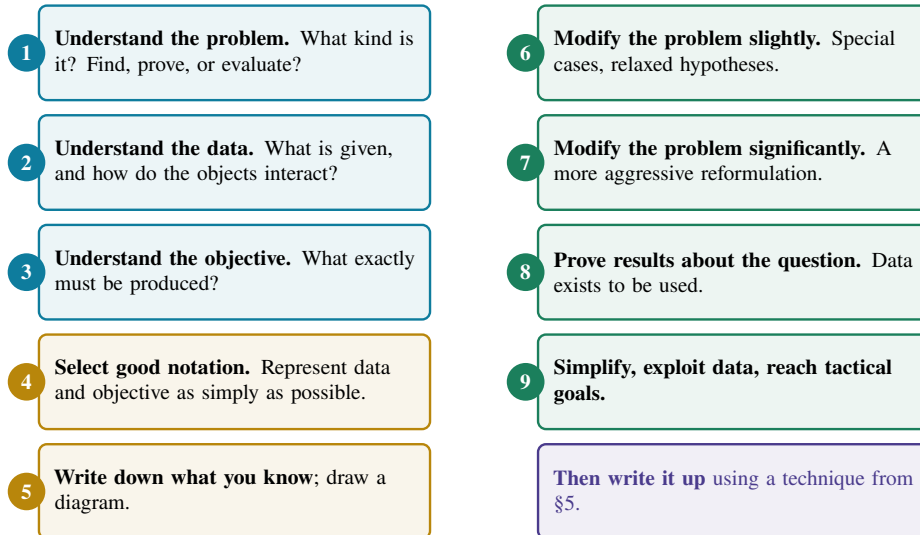


Figure 2. Tao's sequence, from *Solving Mathematical Problems* ch. 1. Steps 1–5 are setup; 6–9 are attack. **Read Tao after you have been stuck**, not before — the advice only lands once you have felt the need for it.

Diedrichs & Lovett's honest warning, §2.3.6

*"There is **no universal strategy** to prove or disprove any given mathematical statement. One of the challenges of advanced mathematics is that the work becomes less and less algorithmic the further one advances. Proving mathematical statements **requires creativity**. . . Nonetheless, there are a few common things to try."*

Their two suggestions: **modify an example or other proof**, and **try a few examples** (with the faulty-generalization caveat from §3).

And the part nobody tells beginners — D&L §2.3.6

*"Though it may take considerable work to find a proof of a statement, **we do not include in the proof all of the ideas, mental movements, dead ends, or examples that led up to our proof.**"*

This is why published proofs look effortless and yours will not. **The finished proof is not a record of how you found it.** Expect the search to be messy; write the result clean.

Worked: how §9's theorem was actually found

Using Tao's numbering, on the theorem $\text{Tr}(T^N) = \lambda_+^N + \lambda_-^N$:

step	what happened
1–3	Objective: a formula for $\text{Tr}(T^N)$. Data: T is 2×2 , real, symmetric.
4	Notation: call the eigenvalues $\lambda_{\pm}$ rather than carrying $2 \cosh \beta$ around.
5	Wrote T out; noticed the symmetry immediately.
6	Modified slightly: tried $N = 1, 2$ by hand. $\text{Tr}(T) = \lambda_+ + \lambda_-$ — suggestive.
8	Proved a result about the question: T symmetric $\Rightarrow$ orthonormal eigenbasis (Axler 7.29). This was the key move.
9	Simplified: in the eigenbasis T^N is trivially diagonal; trace is basis-independent (10.A). Done.

The write-up in §7 is four lines. The search was not.

5 The six techniques, cell by cell

Each technique below gives: its **shape** (the template you fill in), the **book sections** that drill it, the **notebook cell** that demonstrates it, and **when to reach for it**.

Technique	Books	Cell	Reach for it when...
Direct proof	D&L §2.2 “Direct proofs” · Devlin §3.3	CELL 7	the hypothesis gives you something to compute with
Proof by cases	D&L §2.3.1 · Devlin §3.3	CELL 9	the objects split naturally into finitely many kinds
Counterexample	D&L §1.5, §2.2 · Devlin §2.4	CELL 11	you suspect the claim is <i>false</i> . Try this first
Contrapositive	D&L §2.2 “proof by contrapositive”	CELL 13	the negation of the conclusion is easier to work with
Contradiction	D&L §2.3 · Devlin §3.2	CELL 15	the claim says something <i>cannot</i> happen
Induction	D&L §4.5 · Devlin §3.5	CELL 17	the claim is indexed by $n \in \mathbb{N}$

5.1 Technique 1 — Direct proof [CELL 7](#)

Shape

To prove $p \Rightarrow q$: assume p . Apply definitions and known theorems. Arrive at q .

D&L §2.2, in their words

“In a direct proof of a conditional statement $p \rightarrow q$, we assume that p is true, use rules of inference, when needed include as premises other propositions already proved, and conclude that q must be true.”

The notebook’s example. *The Ising transfer matrix T is symmetric.*

Proof. By definition $T(s, s') = \exp[\beta J s s' + \frac{\beta h}{2}(s + s')]$. Swapping s and s' leaves $s s'$ unchanged (multiplication commutes) and leaves $s + s'$ unchanged (addition commutes). So $T(s, s') = T(s', s)$.

■

What to notice. Four lines. It used **the definition**, then **two known facts**. It stopped the moment it reached the claim. [CELL 7](#) then has Sage print `T.is_symmetric() → True` — which checks one 2×2 matrix, whereas the proof covers any number of spin states.

5.2 Technique 2 — Proof by cases [CELL 9](#)

Shape

Split all possibilities into finitely many cases that **cover everything**. Prove each separately.

D&L §2.3.1 — including a warning about taste

*“A proof by cases is a proof of a universal statement that subdivides the universal set into a small number of subsets (cases) and provides a proof for each case separately. The adjective ‘small’ is a value judgment. A proof involving 20 cases... would seem **inelegant** by most mathematicians.”*

The notebook proves $\lambda_+ \geq \lambda_-$ by splitting $\beta \geq 0$ and $\beta < 0$ — **and then points out the cases were unnecessary**, since $\lambda_+ - \lambda_- = 2e^{-\beta} > 0$ for every real β . That is deliberate: *look for the one-line argument before you split.*

5.3 Technique 3 — Counterexample CELL 11

Shape

To disprove $\forall x P(x)$: produce **one** x for which $P(x)$ fails. Nothing else is required.

This is the cheapest technique in mathematics and the notebook makes you use it first. Three claims; two die.

Claim	Verdict
1 every diagonalisable matrix has orthogonal eigenvectors	FALSE. $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ is diagonalisable; its eigenvectors $(1, 1)$ and $(1, 0)$ have dot product 1
2 every real symmetric matrix has distinct eigenvalues	FALSE. The identity has eigenvalues 1, 1
3 every real symmetric matrix is diagonalisable	TRUE — and no search can establish it. This is Axler 7.29, and it takes six pages

Why claim 1 is the one that matters to you

It is exactly why Axler 7.29 says **orthonormal** and not merely “a basis of eigenvectors”. Orthogonality is an extra gift that symmetry buys — and it is what makes the trace argument in §7 work without cross terms.

Claims 1 and 2 died to a single 2×2 matrix each. Claim 3 survived every search — and surviving a search is not a proof. That asymmetry is §3 of this document, made concrete.

5.4 Technique 4 — Contrapositive CELL 13

Shape

$p \Rightarrow q$ is *logically identical* to $\neg q \Rightarrow \neg p$. Prove whichever is easier.

D&L §2.2

“A proof by contrapositive. . . is a direct proof of the contrapositive $\neg q \rightarrow \neg p$. Since the contrapositive of a conditional statement is equivalent to the original. . . proving $\neg q \rightarrow \neg p$ proves $p \rightarrow q$.”

And on style: “The signal ‘We prove the contrapositive’ signals to the reader our proof strategy.” — announce your technique. That is the communication half of Devlin’s definition.

CELL 13 prints the two truth-table columns side by side; they agree in all four rows.

5.5 Technique 5 — Contradiction CELL 15

Shape

Assume the **opposite** of what you want. Derive something impossible. Conclude the assumption was false.

The notebook’s example. *The 1D Ising chain has no phase transition at finite $\beta > 0$.*

Proof. A phase transition means $f = -\frac{1}{\beta} \ln(2 \cosh \beta)$ is non-analytic at some finite $\beta_c > 0$. Suppose so. The only way $\ln(2 \cosh \beta)$ can fail to be analytic is if $2 \cosh \beta_c = 0$. But $\cosh \beta = \frac{1}{2}(e^\beta + e^{-\beta})$ is a sum of two strictly positive numbers, so $\cosh \beta_c > 0$. Contradiction. ■

CELL 15 corroborates with `solve(2*cosh(beta)==0, beta) → []` — an empty list, exactly as the argument predicted.

5.6 Technique 6 — Induction CELL 17

Shape — the one that beats infinity

To prove $P(n)$ for all $n \geq 1$:

1. Base case: prove $P(1)$.

2. Inductive step: assume $P(k)$ for arbitrary k ; prove $P(k + 1)$ follows.

Then $P(1)$ gives $P(2)$ gives $P(3)$... forever. **Finite work, infinitely many claims.**

The lemma your Ising notebook assumed without proof. If D is diagonal with entries d_1, d_2 then D^n is diagonal with entries d_1^n, d_2^n .

Base case $n = 1$. $D^1 = D$. ✓

Inductive step. Assume $D^k = \text{diag}(d_1^k, d_2^k)$. Then $D^{k+1} = D^k \cdot D = \text{diag}(d_1^k d_1, d_2^k d_2) = \text{diag}(d_1^{k+1}, d_2^{k+1})$, using only that diagonal matrices multiply entrywise. ✓ ■

CELL 17 is where the lesson lands

The cell checks $n = 1 \dots 20$ and prints all diagonal? True, then says:

```
*** READ THIS CAREFULLY ***
Sage verified 20 cases. The lemma claims INFINITELY many.
The 20 checks are evidence that we did not misstate it.
The PROOF is the inductive step above - and only that.
```

This is the single most important cell in the notebook. Everything else is scaffolding for this distinction.

6 The CAS boundary — what Sage cannot do

CELL 21

Sage CAN

- **Falsify fast.** One counterexample kills a conjecture — the cheapest mathematics there is [CELL 11](#)
- **Verify instances.** $ns=3, 5, 8$ before you attempt the general case
- **Do the algebra inside a proof.** `exponentialize()` closed a hyperbolic identity you would otherwise grind by hand
- **Suggest the closed form.** Compute, spot the pattern, then prove it

Sage CANNOT

- **Prove $\forall n$.** Twenty checks are twenty facts [CELL 17](#)
- **Tell you why.** True is not an explanation
- **Choose the right lemma.** Nothing in Sage suggests “invoke the spectral theorem here”
- **Be trusted when it says False.** See below — this one has already bitten you

6.1 The trap: a CAS that says False about a true statement

[CELL 21](#) runs one expression through four simplification methods. The expression is **identically zero**.

```
the expression: -(e^(2*beta) + 1)*e^(-beta) + 2*cosh(beta)
bool(expr == 0) -> False
bool(expr.simplify_full() == 0) -> False
bool(expr.canonicalize_radical()==0) -> False
bool(expr.exponentialize()... == 0) -> True

substitute any number and see:
beta=0.3: 0.000e-12
beta=1.0: -4.441e-16
beta=2.5: -1.776e-15
```

The moral, stated precisely

A CAS False is not a disproof. It is a failure to simplify.

Three of four methods failed on a true identity. Had you trusted the first, you would have concluded $\lambda_+ \neq 2 \cosh \beta$ — which is false, and would have wrecked every derivation downstream.

The converse is weaker than it looks too: a True from a sound method verifies *exactly what you asked*, over *exactly the domain you declared*. Note that the notebook declares `var('beta', domain='real')`. Drop that and the simplifications change.

Thurston saw this coming, in 1994

On the four-colour theorem, proved by exhaustive computer search:

“it evoked much controversy. I interpret the controversy as having little to do with doubt people had as to the veracity of the theorem or the correctness of the proof. Rather, it reflected a continuing desire for human understanding of a proof, in addition to knowledge that the theorem is true.”

That is the distinction this whole notebook is built around. Sage can give you knowledge that a statement is true for the cases you checked. Understanding is a different commodity, and you have to make it yourself.

7 The payoff — the theorem, assembled

CELL 19

Everything in §5 was preparation for this. The notebook's §9 states and proves the theorem your Ising notebooks had only checked.

Theorem

For the 1D Ising transfer matrix T and every $N \geq 1$,

$$\text{Tr}(T^N) = \lambda_+^N + \lambda_-^N.$$

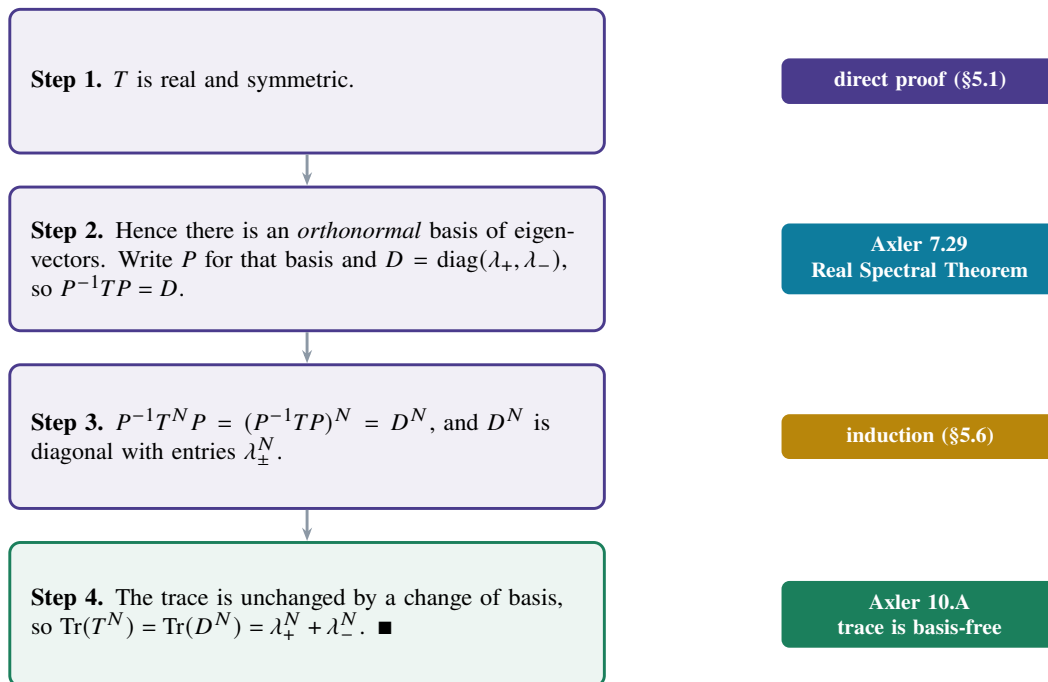


Figure 3. Four steps, each licensed by something named. **That is what a proof looks like.** It holds for $N = 10^{100}$ with no further work — which no amount of computation could give you.

The gap left deliberately open

Step 3 used $(P^{-1}AP)^N = P^{-1}A^N P$ **without proving it**. That is not an oversight — it is notebook **exercise 6**, and it is your first genuine induction.

Try it before reading on. Base case $N = 1$ is immediate; for the step, insert $PP^{-1} = I$ in the middle and watch it telescope.

One theorem, three languages — CELL 22 , the notebook's §11

	statement	lives in
Pure	a self-adjoint operator on a finite-dimensional real inner-product space has an orthonormal eigenbasis	Axler 7.29
Applied	the free energy is $-\frac{1}{\beta} \ln \lambda_+$ and $\xi = 1/\ln(\lambda_+/\lambda_-)$	Feynman ch. 5; PBoC2 ch. 6
Computational	<code>T0.eigenmatrix_right()</code> returns (D, P) with P orthogonal	Sage

The **pure** statement licenses the applied one. The **computational** one only checks instances. Losing track of which you are doing is the commonest error in computational science.

8 Complete index — all 22 cells

Twelve markdown cells carry the mathematics; ten code cells carry the checks. **The mathematics is in the markdown.** If you read only the code you will learn Sage and no proof.

#	type	nb §	what it does	what to take from it
1	MD	Title, §0	Defines proof; lists the three books used throughout	The two-line definition of a proof. Memorise it
2	MD	§1	Section heading	—
3	CODE	§1	<code>version()</code> ; declares β real	Never name a variable N — $N()$ is Sage's numeric evaluator
4	MD	§2	Implication $p \Rightarrow q$; the two beginner errors	False p makes $p \Rightarrow q$ true; converse $\neq$ original
5	CODE	§2	Truth table from $(\text{not } p) \text{ or } q$	That expression <i>is</i> the definition of $\Rightarrow$
6	MD	§2	Quantifiers $\forall, \exists$ and their negation	The asymmetry: disproving $\forall$ costs one example; proving it costs an argument
7	CODE	§3	Builds T in SR; <code>T.is_symmetric()</code>	Sage <i>checked</i> a 2×2 ; the proof explains <i>why</i> , for any number of states
8	MD	§4	Proof by cases: $\lambda_+ \geq \lambda_-$	And the honest footnote — the cases were unnecessary. Look for the one-liner first
9	CODE	§4	<code>exponentialize().simplify_full()</code> gives $2e^{-\beta}$	Your first sight of the method that closes hyperbolic identities
10	MD	§5	Three claims, two false. <i>Think before running</i>	The pause is the exercise. Predict, then check
11	CODE	§5	$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ and I kill claims 1–2; claim 3 survives	A surviving claim is not a proved claim
12	MD	§5–6	The lesson; then contrapositive	Why Axler says <i>orthonormal</i> , not just “a basis of eigenvectors”
13	CODE	§6	Truth table: $p \Rightarrow q$ vs $\neg q \Rightarrow \neg p$	Identical in all four rows $\Rightarrow$ interchangeable. Same statement, easier door
14	MD	§7	Contradiction: no 1D phase transition at finite β	Assume the opposite; derive $2 \cosh \beta_c = 0$; impossible
15	CODE	§7	<code>solve(2*cosh(beta)==0)</code> returns <code>[]</code> ; four derivatives finite	Empty solution set matches the argument — <i>corroborates</i> , does not prove
16	MD	§8	Induction; the diagonal-power lemma proved	Base case + inductive step = infinitely many claims, finite work
17	CODE	§8	Checks $n = 1 \dots 20$, then says so in capitals	20 checks are 20 facts. The proof is the step, and only the step
18	MD	§9	The theorem , four steps, each licensed	The payoff. Holds for $N = 10^{100}$
19	CODE	§9	Diagonalises T_0 ; <code>proves_zero</code> on $ns = 1, 2, 3, 7, 12$	<code>eigenmatrix_right()</code> returns (D, P) — diagonal first. Getting this backwards silently breaks the check
20	MD	§10	“Where the CAS will lie to you”	Framing for the cell below
21	CODE	§10	Four methods on an identically-zero expression	Three say False. A CAS False is a failure to simplify, not a disproof
22	MD	§11–12	Pure/applied/computational table; 8 exercises	Exercise 6 is the induction left open in §9. Do it

Read in this order

First pass: cells 1, 4, 6, 8, 10, 12, 14, 16, 18, 22 — the markdown alone. That is the mathematics, and it is about forty minutes.

Second pass: run the code cells and check each output against what the markdown predicted. Where they differ, the markdown is right and you mis-read the code.

Third pass: do the exercises in cell 22 with the notebook *closed*.

9 A four-week route through the eight books

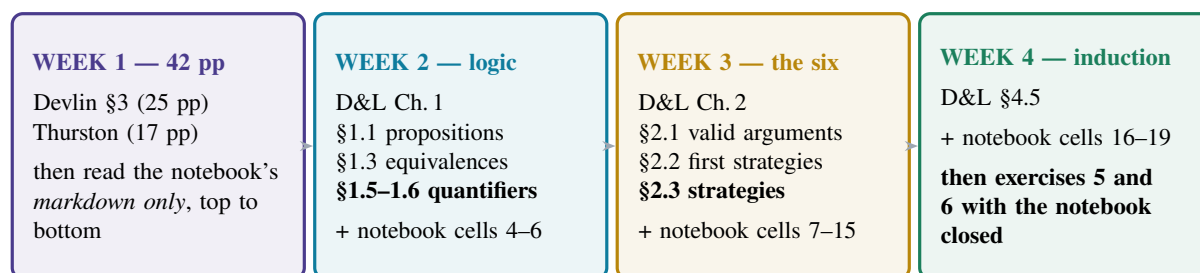


Figure 4. You do not read 2,164 pages. Week 1 is **42 pages**; the rest is drill. Diedrichs & Lovett is a gym, not a novel — read one technique, do ten exercises, move on.

9.1 What each book is actually for

book	pp	folder	what it is for, and when
Devlin , <i>Introduction to Mathematical Thinking</i>	117	01_Start_Here	Start here, this week. The shortest complete route into proof that exists. §3 is 25 pages and covers every technique in the notebook
Thurston , <i>On Proof and Progress</i>	17	01_Start_Here	Read in one sitting, early. Not a textbook — a Fields medallist asking what proof is <i>for</i> . It will change what you think you are doing
Diedrichs & Lovett , <i>Transition to Advanced Mathematics</i>	552	02_Main_Reference	Your drill manual. Ch. 2 names exactly the six techniques; hundreds of exercises with answers. Weeks 2–4
Stewart & Tall , <i>Foundations of Mathematics</i>	409	03_Second_Bridge	The second voice. When D&L's explanation does not land, read theirs. Same material, different telling
Tao , <i>Solving Mathematical Problems</i>	116	04_Problem_Strategy	Strategy, not technique. Read <i>after</i> you have attempted and failed at real problems — see §4 of this tutorial
Halmos , <i>Naive Set Theory</i>	112	05_Logic_and_Sets	Terse and beautiful; read <i>after</i> Devlin, for elegance
O'Leary , <i>First Course in Mathematical Logic and Set Theory</i>	466	05_Logic_and_Sets	Full reference for formal logic. Not a first read
Radozycki , <i>Solving Problems in Mathematical Analysis I</i>	375	06_Worked_Problems	Read the solutions as models of written argument. Most books show results; this one shows the <i>writing</i>
8 books	2,164	6 folders	all MD5-verified byte-identical to their originals

The one habit that matters

After every proof you read, close the book and **write it out from memory**. Where you stall is exactly where you did not understand it — and you will not discover that any other way. D&L §2.3.6 puts it plainly: “we do not include in the proof all of the ideas, mental movements, dead ends, or examples that led up to our proof.” The written proof hides the search. Redoing it is how you find the search again.

10 Verification ledger — what was checked, and how

Every claim of fact in this tutorial was checked on your machine on **19 September 2026**. Where a check could not be made, it says so.

claim	how checked	result
Notebook has 22 cells, 10 of them code	<code>json.load</code> over the <code>.ipynb</code>	22 cells, 10 code, 12 markdown
Notebook runs clean under Sage	scan every cell's outputs for <code>output_type=='error'</code>	0 errors , 10 cells executed
Kernel is Sage 10.9	<code>metadata.kernelspec</code>	SageMath 10.9 / <code>sagemath-10.9</code>
Eight proof books present	find over <code>Maths_Proof_Skill/</code>	8 PDFs, 6 folders, 22 MB
Copies are byte-identical to sources	MD5 of every copy vs. its original	8/8 match ; originals untouched
Quotations in §2–§4	read from <i>your</i> PDFs, not from memory	Devlin §3.1; D&L Def. 2.1.15 and §2.3.6; Thurston; Tao
Cell numbers cited in §5–§8	re-extracted from the notebook and cross-checked	corrected once during drafting — see note below

One correction made while writing this document

An early draft cited the CAS-failure demonstration as cell 19. Re-extracting the cell list showed cell 19 is the *theorem verification* and the CAS demonstration is **cell 21**. Every **CELL *n*** tag in this document was then re-derived from the notebook file rather than from the draft. **This is the same discipline the document argues for:** a claim that was only plausible got checked against the source, and it was wrong.

10.1 Three things this tutorial deliberately does not claim

1. **That you can now write proofs.** You can now *read* six techniques and recognise which one a proof is using. Writing them is weeks 2–4, and it is exercises, not reading.
2. **That the notebook's Sage checks prove anything.** They do not, and the notebook says so in capitals at cell 17. Every proof in it is in the markdown.
3. **That the four-week route is enough for Axler.** It gets you to the point where Axler §7.B is readable. *Readable*, not easy — that is Week 1 of the twelve-week partition-function plan, and a different document.

The one sentence to keep

Computation tells you *that* something is true for the cases you ran.
A proof tells you *why* it is true, for all the cases you will never run.

Devlin: “a proof of a statement must explain why that statement is true.”

Thurston: “a continuing desire for human understanding of a proof, in addition to knowledge that the theorem is true.”